Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/18/2017
PI Name:	Clark, Torin K. Ph.D.		
Project Title:	Predicting Sensorimotor Adaptation to Altered Gravity by Measuring Vestibular Perceptual Thresholds		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80303-7813	**Congressional District:**	2
Comments:	NOTE: PI moved to University of Colorado after NSBRI Postdoctoral Fellowship concluded in late 2015 (Ed., 9/1/17)		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 NSBRI-RFA-13-01 Postdoctoral Fellowships
Start Date:	10/01/2013	**End Date:**	09/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Merfeld, Daniel (MENTOR/ Massachusetts Eye and Ear Infirmary)		
Grant/Contract No.:	NCC 9-58-PF03502		
Performance Goal No.:			
Performance Goal Text:			
	POSTDOCTORAL FELLOWSHIP During and shortly after gravity transitions astronauts suffer from sensorimotor impairment which may impact critical mission tasks, particularly vehicle control and landing. While all astronauts eventually adapt to the altered gravity environments reducing sensorimotor impairment, there appear to be substantial individual differences in the severity of impairment and how long is required for adaptation, which are poorly understood. We have proposed a novel hypothesis to explain individual differences in sensorimotor adaptability: sensory noise, which we can assay using threshold measures—most relevantly vestibular perceptual thresholds—is a limiting factor in sensorimotor adaptation. We hypothesize individuals with higher noise (higher vestibular perceptual thresholds) will require a more sustained sensory conflict before adaptation is appropriate and thus take longer to adapt to an altered gravity environment.		

Task Description:	In our primary experiment, we test this hypothesis by 1) measuring an individual's vestibular perceptual thresholds (how small of a self-motion can be reliably perceived), then 2) quantifying that individual's adaptability to an altered gravity environment (adaptation in orientation perception to a hyper-gravity test-bed using a centrifuge), and finally 3) testing for a correlation between thresholds and adaptation rate. Our preliminary data suggest there is indeed a strong relationship between thresholds and adaptation rate and our ongoing experiments are aimed at further evaluating our hypothesis. We aim to deliver a practical, operationally-relevant pre-flight predictor of sensorimotor adaptability to NASA. In addition to being well correlated with adaptation, in order for thresholds to be an operationally-relevant predictor they must A) be highly stable within an individual over time, since the pre-flight test may be weeks to months (or even years) before a critical gravity transition during flight and B) must be highly efficient (e.g., short testing time). To address item A, we have conducted an extensive study repeatedly measuring thresholds in a small population of subjects regularly for nearly one year and have found a remarkable consistency within each individual (contrast that with the ~10x variability in thresholds between individuals). To address item B, we have begun development on a confidence task and fitting method for threshold measurement, which our simulations and preliminary data suggest may reduce the required testing time by ~5x. Future research will continue to investigate the relationship between an individual's sensory noise level and their ability to adapt to altered gravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are several Earth-based impacts of the current research project. First, while predicting an individual's capacity for sensorimotor adaptation is important for astronauts on orbit, it is also critical for sensorimotor rehabilitation patients here on Earth. Being able to predict a patient's capacity for sensorimotor adaptation may allow individualized rehabilitation plans. Second, our advances in measuring vestibular perceptual thresholds may lead to a powerful diagnostic tool in the clinic. Current practices rely almost entirely on vestibular ocular reflex (VOR) responses to diagnose patients complaining of dizziness or vertigo. However, since symptoms (dizziness or vertigo) are illusory perceptions these reflex response tests may leave patients undiagnosed. Using vestibular perceptual thresholds as part of the test battery may provide improved diagnostics in the clinic here on Earth.
Task Progress:	In the last year we have made substantial technical progress. First, we have modified our Eccentric Rotator (XR) centrifuge device for our altered gravity motion paradigm. This has included technical development as well as extensive safety checks. We have implemented a somatosensory (as referred to as haptic) task for reporting roll tilt perception in which the subject aligns a bar with their perceived horizontal. We have also implemented several version of a subjective visual vertical (SVV) task in which an illuminated line is presented on a computer monitor in front of the subject and they adjust the line until it appears vertical. In addition to these tasks for orientation perception we have developed and implemented a manual control task in which the subject attempts to null out a disturbance and keep his/her chair aligned with upright. We have tested these tasks in many human subjects in various paradigms in which the centripetal acceleration magnitude is equal to 1 G (baseline), greater than 1 G (hyper-G), or less than 1 G (hypo-G) to study adaptation in the responses. We have developed and implemented a novel methodology for measuring vestibular perceptual thresholds. In the standard testing protocol, on each trial the subject moves to the left or right (randomly determined) and after the motion report whether they thought they moved to the left or right. We estimate the threshold as the smallest motions which can reliably be perceived. In our novel method, subjects not only report whether they felt they moved left or right (same as the standard task), but also their confidence in that selection. Confidence is reported on a scale between 50% and 100% using increments of 1%. The binary and confidence reports are made using a custom iPad interface. Our laboratory (Dr. Yongwoo Yi) has developed a novel fitting approach to determine a threshold estimate from data from this task. Our simulations and preliminary results suggest this novel confidence task will yield similar threshold estimates in ~5x as few trials, dramatically reducing testing time. This is important not only for research but also for future testing of astronauts, given their limited availability. With the successful progress of the project, we have produced several publications. In particular, we have published 3 journal articles and have 2 more currently under review. In addition, we have 2 more journal articles in preparation for submission that will be submitted shortly (one on the consistency of vestibular thresholds and one on lapse detection for threshold measurement). Finally, there are several conference papers, abstracts, posters, and presentations which are attributable to this project.
Bibliography Type:	Description: (Last Updated: 11/28/2025)
Abstracts for Journals and Proceedings	Clark TK, Galvan-Garza RC, Bermudez Rey MC, Yi Y, Merfeld DM. "Perceptual Noise and Sensorimotor Adaptation." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Abstracts for Journals and Proceedings	Clark TK, Newman MC, Oman CM, Merfeld DM, Young LR. "Modeling Human Orientation Perception in Altered Gravity." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Abstracts for Journals and Proceedings	Diaz A, Beckers NWM, Clark TK, Sherwood D, Oman C, Young LR, Karmali F. "Development of a Countermeasure to Enhance Sensorimotor Adaptation to Altered Gravity Levels." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Abstracts for Journals and Proceedings	Galvan RC, Clark TK, Merfeld DM, Bloomberg JJ, Mulavara AP, Oman CM. "Improving Sensorimotor Function using Stochastic Vestibular Stimulation." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015

Abstracts for Journals and Proceedings	Hackler AS, Deymeier-Black A, Clark TK, Lawley J, Simon J, Vokhari R, LaPelusa M, McNeel R. "Innovation by a New NSBRI Generation." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. , Jan-2015
Articles in Peer-reviewed Journals	Clark TK, Newman MC, Merfeld DM, Oman CM, Young LR. "Human manual control performance in hyper-gravity." Exp Brain Res. 2015 May;233(5):1409-20. Epub 2015 Feb 5. https://doi.org/10.1007/s00221-015-4215-y ; PubMed PMID: 25651980 , May-2015
Articles in Peer-reviewed Journals	Clark TK, Newman MC, Oman CM, Merfeld DM, Young LR. "Human perceptual overestimation of whole body roll tilt in hypergravity." J Neurophysiol. 2015 Apr 1;113(7):2062-77. Epub 2014 Dec 24. https://doi.org/10.1152/jn.00095.2014 ; PubMed PMID: 25540216; PubMed Central PMCID: PMC4416546 , Apr-2015
Articles in Peer-reviewed Journals	Clark TK, Newman MC, Oman CM, Merfeld DM, Young LR. "Modeling human perception of orientation in altered gravity." Front Syst Neurosci. 2015 May 5;9:68. eCollection 2015. https://doi.org/10.3389/fnsys.2015.00068 ; PubMed PMID: 25999822; PubMed Central PMCID: PMC4419856 , May-2015